

An Auditable Market-Microstructure Framework for Spoofing-Compatible Sequence Detection

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Abstract

We develop an auditable microstructure framework for screening order-book sequences that are compatible with spoofing. Legitimate liquidity provision remains an explicit alternative explanation. The framework reconstructs the visible limit order book and consolidates related executions into clusters. A cluster can be anchored either to the participant’s resting order being executed or to the participant trading aggressively. We attribute orders through the original client code when it is available and otherwise through the firm identifier, while preserving the identity level. Each cluster is linked to same-actor orders posted on the opposite side during the preceding 90 seconds and withdrawn within 2 seconds after the cluster. An empirical top-10 depth kernel weights visible liquidity by book rank. We apply the framework to three Italian equity samples and report mechanical withdrawal and price-response diagnostics. The resulting conjunction supports reproducible surveillance triage. It does not constitute a statistical test, identify manipulative intent, estimate causal effects, or measure the prevalence of spoofing.

Keywords. Spoofing detection; market manipulation; limit order book; high-frequency trading; anomaly detection; market maker; market microstructure

1 Introduction

Electronic continuous double auctions organize price discovery through a limit order book (LOB). The LOB records the visible bid and ask orders submitted by market participants and changes whenever an order is entered, amended, executed, or cancelled. Algorithmic and high-frequency trading have increased the speed of these updates and can improve short-term liquidity. The same speed also allows a participant to create and remove large displayed positions before other traders can assess the full sequence.

Spoofing exploits this sequencing problem. In the stylized mechanism, a participant submits a large order on one side of the LOB without intending to execute it. The displayed quantity can

create a misleading signal of supply or demand and can alter order-book imbalance or microprice indicators. Other participants may respond by changing their quotes or submitting aggressive orders. The participant can then execute a smaller order on the opposite side at a more favorable price and withdraw the large displayed order. The Market Abuse Regulation (MAR) prohibits orders and behavior that create false or misleading signals about the supply of, demand for, or price of a financial instrument [3].

The observable actions in this mechanism are not unique to spoofing. Liquidity providers also submit, amend, and cancel orders at high frequency, and they may have high order-to-trade ratios. They remove quotes to update prices, control inventory, or limit adverse selection. Static size thresholds, cancellation ratios, and volumetric triggers therefore cannot by themselves distinguish manipulative conduct from legitimate liquidity provision. Such rules can produce false positives and cannot establish the participant’s intent.

We address this surveillance problem by reconstructing participant-level order sequences rather than classifying isolated messages. The framework combines five observable elements: a recent opposite-side liquidity profile, an execution by the same actor, a rapid withdrawal of that profile, a price movement favorable to the execution, and subsequent price reversion. The execution can be passive or aggressive. Actor identity is a separate data problem: we use the original client code when present and the firm identifier only as a fallback. Related execution messages are consolidated into clusters, and each physical cancellation is assigned to at most one cluster. The output is an auditable set of descriptive screening diagnostics for analyst review, not a statistical validation, a label of manipulation, or an estimate of causal price impact.

2 Market Making Versus Spoofing

The screening model starts from a stylized contrast between market making and spoofing. Both activities can generate many order messages and cancellations. They differ, however, in the role that displayed liquidity plays in the participant’s objective. This contrast does not imply that every market maker quotes symmetrically or that every spoofing episode follows one fixed pattern. It identifies observable features that can guide surveillance.

2.1 The Market Maker’s Problem

A market maker supplies liquidity by posting resting orders on the bid and ask sides of the book. In a stylized inventory-risk benchmark, the dealer seeks to earn the bid–ask spread while controlling inventory risk and exposure to adverse selection. Optimal quotes therefore depend on both order-arrival rates and the dealer’s inventory.

Stochastic-control models often express this problem through a Hamilton–Jacobi–Bellman equation. The dealer chooses bid and ask quotes that balance execution revenue against inventory risk. For example, a long inventory exposes the dealer to a price decline. The dealer can respond by shifting both quotes downward: a lower ask encourages purchases from the dealer, while a lower bid discourages further sales to the dealer. The quote adjustment depends on the mid-price, the spread required for compensation, and the penalty for inventory away from the target.

Within this benchmark, frequent cancellation is a risk-management action. The market maker cancels and reposts orders as the mid-price changes, as adverse-selection risk changes, or as inventory moves away from its target. The dealer therefore tends to maintain a two-sided presence near the top of the book, although temporary asymmetries can arise from inventory management. Expected profits come primarily from spread capture rather than from deliberately inducing a directional change in the mid-price.

2.2 The Spoofer’s Problem

The stylized spoofer instead seeks to improve the price of a directional execution. The participant displays a large order on one side of the book and submits a smaller order on the opposite side. The large displayed order creates an asymmetric depth profile. Algorithms or traders that use bid–ask volume imbalance may interpret this profile as evidence of short-term price pressure and adjust their orders. The participant can benefit if the resulting price movement improves the execution of the smaller opposite-side order.

The displayed order creates a trade-off between influence and execution risk. An order near the best quote is prominent in the book but is also more likely to execute. An order posted at Level 2, Level 3, or deeper has a buffer against immediate execution while remaining visible in multilevel order-book data. A participant implementing the stylized strategy may therefore distribute the large profile across deeper levels rather than place all of it at the touch.

After the smaller order executes, the displayed opposite-side profile no longer serves the hypothesized strategy and continues to expose the participant to unwanted execution. Under this mechanism, the participant therefore has an incentive to cancel it quickly. This timing motivates the diagnostic contrast with ordinary quote updates: the candidate withdrawal follows the opposite-side execution rather than only changes in fair value, inventory, or adverse-selection risk. [Table 1](#) summarizes these stylized incentives; none is directly observed as intent.

3 Existing Approaches

Spoofing-detection methods differ mainly in the object they classify. Rule-based methods reconstruct a stylized episode from order-book mechanics. Statistical methods monitor unusual states of order flow. Sequence and graph models learn temporal or relational patterns from event data. This distinction is more informative for our setting than a broad split between statistical and machine-learning methods. [Figure 1](#) summarizes the method families, which can overlap in practice.

Early empirical studies operationalize spoofing-surveillance screens through interpretable order-book conditions. Using account-level Korea Exchange data, [\[7\]](#) identify large orders placed far from the best quote, followed by opposite-side trading and subsequent withdrawal. Related surveillance rules have been applied to Brazilian equities [\[9\]](#), FX spot markets [\[11\]](#), cross-market FX activity [\[10\]](#), and Taiwan derivatives [\[1\]](#). These studies connect a screen to observable features such as order size, posting distance, cancellation timing, opposite-side execution, and short-lived price impact. However, the reported implementations use venue-specific thresholds, and their reconstruction can require privileged message data or account identifiers.

Table 1: Stylized differences used to construct the surveillance diagnostics. These differences describe economic benchmarks, not observed intent or necessary conditions for either activity.

Microstructure Feature	Legitimate Market Maker	Hypothesized Spoofing Mechanism
Primary objective	Earn the bid–ask spread while controlling inventory and adverse-selection risk.	Improve a directional execution through a displayed order-book signal.
Order-book imbalance	Usually maintains two-sided liquidity, with asymmetry arising from inventory and risk management.	Uses a large, short-lived asymmetry as the candidate posture.
Cancellation motive	Update quotes to track fair value, control inventory, or limit adverse selection.	Remove the displayed posture after the smaller opposite-side execution.
Posting-distance trade-off	Quote near the touch to increase execution probability and earn the spread.	Remain visible while reducing the risk that the displayed posture executes.
Source of expected gain	Spread capture, without intentionally moving the mid-price in a chosen direction.	A favorable price change for the smaller order on the opposite side.

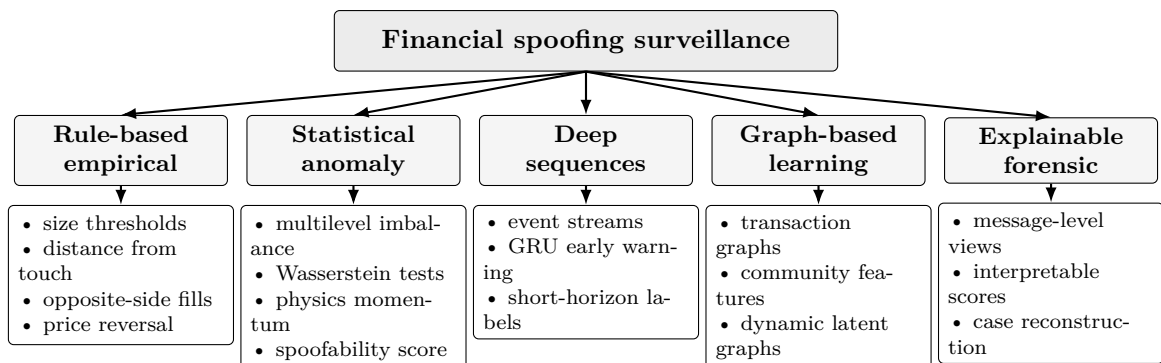


Figure 1: Method families in the spoofing-detection literature. The categories can overlap; each column identifies the main object or representation used by a strand of work.

A second line of work monitors statistical anomalies in the order book. [13] derive spoofing incentives from multilevel imbalance and propose a Wasserstein-distance statistic for real-time monitoring. [8] represent the order book as a statistical-physics system and detect abnormal momentum in cryptocurrency markets. Explainable forensic methods partly overlap with this family. Message-level visualization can reconstruct prosecuted episodes [2]. Interpretable probabilistic neural networks instead estimate how an isolated limit order changes the conditional distribution of future prices and the expected gain from a hypothetical manipulation strategy in centralized cryptocurrency markets [4]. The latter object measures order-level *spoofability*; it does not establish whether one participant generated the candidate posture, the smaller execution-side order, and the withdrawal. Our framework addresses this complementary attribution problem by linking these observed actions through anonymized actor identifiers at the client or firm level.

Machine-learning methods add temporal or relational structure. [14] formulate spoofing as an early-warning problem and train a gated recurrent unit model on cryptocurrency order-book event streams. Graph-based methods move the unit of analysis from isolated orders to networks of traders and transactions. This representation can capture coordinated spoofing patterns [5, 6, 15]. Reported models combine temporal transaction graphs, community features, and dynamic latent representations. Their practical performance remains difficult to compare because open labels and chronological cross-market validation are limited.

These studies imply two requirements for the present surveillance setting. First, the features must retain their microstructure interpretation. The screen must record candidate volume, posting distance from the touch, cancellation timing, and the relation to an opposite-side execution. Second, evaluation must reflect scarce and noisy enforcement labels, rare positive cases, and the analyst cost of false positives. The model below therefore preserves the auditability of rule-based surveillance while using participant-level order sequences from proprietary event data.

4 Data

We use three CONSOB limit-order-book message samples. The Risanamento sample covers June–November 2024, the Nexi sample covers 23 April 2024, and the Ferrari sample covers 13 June 2024. These samples differ in duration and participant composition, so cross-instrument differences are descriptive rather than controlled comparisons.

For each instrument, we reconstruct the visible order book from order-level events and infer the tick size from the reconstructed best quotes. We compute episode-conditioned metrics over the first ten book ranks. The main specification uses a 10-second structural post-execution window, a 2-second rapid-withdrawal attribution window, a 2-second reversion horizon measured from each assigned cancellation, and a maximum candidate-order age of 90 seconds.

The original client code is not available on every message. We therefore define one canonical actor key before constructing events. If the original client code is present, it defines the actor; only when it is missing do we use the firm identifier. The two namespaces are kept distinct, and every output records whether attribution is at the client or firm level. Messages missing both identifiers are excluded from actor-attributed analysis; none occurs in the reported samples. Firm fallback is coarser than client attribution and can combine several clients represented by the

same firm. It is therefore an identity fallback, not evidence that those clients are one economic account.

Execution role is defined independently of actor identity. We retain fills marked exclusively passive or exclusively aggressive and reject fills for which the two role flags are both set or both absent. Related messages are consolidated into one execution cluster when they share the trading partition, canonical actor, order identifier, execution role, and side. Passive clusters also require a common resting price; aggressive clusters can span several traded prices under the same order and sweep. Consecutive fills must be no more than 100 milliseconds apart, with no intervening lifecycle boundary. The 100-millisecond threshold is an operational choice, and the reported findings are conditional on it; sensitivity to alternative inter-fill gaps is not reported here. Raw-message identifiers and cluster memberships remain available as provenance, while execution clusters form the analytical unit for event metrics and actor-partition aggregation. The resulting surveillance dossiers support analyst review and are not enforcement labels.

5 Model

This section translates the stylized contrast between market making and spoofing into observable sequence diagnostics. We first represent each participant’s liquidity across the top n book levels. We then measure whether a recent opposite-side profile disappears after the participant’s own execution and whether the associated price path is economically consistent with the hypothesized strategy.

5.1 A Spatio-Temporal Framework for Spoofing Detection

Figure 2 sketches the hypothesized sequence. The detector later translates each state into observable order, execution, and cancellation records rather than inferring the participant’s purpose.

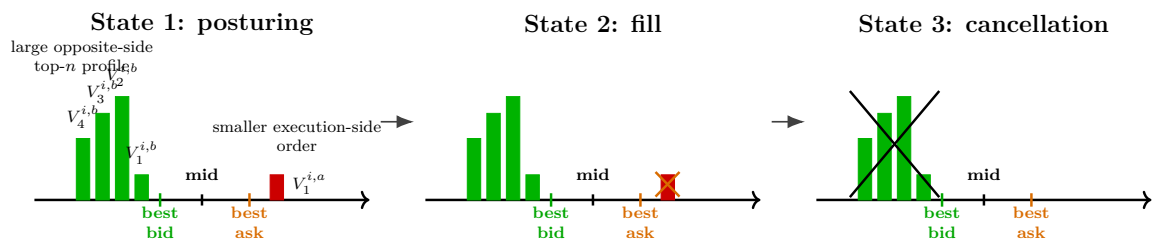


Figure 2: Schematic top- n timeline for the hypothesized spoofing mechanism. The large candidate posture forms an agent-specific depth profile over several bid levels, while the smaller order rests near the opposite best quote. The construction applies symmetrically to candidate buy- and sell-intent episodes. The figure shows the passive execution route; the operational reconstruction also admits an aggressive execution by the same actor.

The method represents a candidate episode as a conditional change in the participant’s LOB footprint. A single order at the best bid or ask does not capture this change. We instead use the participant’s liquidity profile over the first n price levels. This representation includes postures placed at Level 2, Level 3, or across adjacent levels, where displayed liquidity can remain visible while facing less immediate execution risk.

The hypothesized mechanism for participant i has three states:

1. **State 1 (Candidate Posturing):** Shortly before the execution, the participant displays a large liquidity profile on one side of the book, typically away from the touch. The participant also maintains a smaller order near the opposite best quote.
2. **State 2 (Execution):** The large profile is present when the smaller opposite-side order executes. In the hypothesized mechanism, the profile contributes mechanically to short-horizon imbalance statistics before this execution.
3. **State 3 (Withdrawal):** The participant rapidly cancels the opposite-side profile after the execution, ending its exposure to unintended fills of those orders.

State 2 has two observable routes. In the passive route, the actor's resting order executes. In the aggressive route, the actor initiates the trade against resting liquidity. The role flag changes how execution price and quantity are read and how child messages are clustered; it does not change actor identity. In both routes, the candidate posture is sought among the same actor's recent orders on the side opposite to the actor's execution side. An aggressive anchor is therefore not itself evidence of spoofing, just as a passive anchor is not evidence of legitimate market making.

The detector observes the sequence, not the participant's intent or the profile's causal effect on other traders. It therefore requires the asymmetric profile to be recent, present before the fill, and withdrawn conditionally after the fill. The recency restriction prevents long-lived liquidity on the opposite side of a later execution from entering the candidate episode solely because it was already resting in the book.

5.1.1 Top- n Participant-Specific Depth Profiles

We first define the participant-specific depth profile used throughout the model. Let $s \in \{b, a\}$ denote the bid or ask side, and let $k = 1, \dots, n$ denote book rank. Rank $k = 1$ is the best quote on side s . For participant i , define

$$V_k^{i,s}(t) \tag{1}$$

as the participant's displayed volume at time t on side s and rank k . The vector

$$\mathbf{V}_n^{i,s}(t) = (V_1^{i,s}(t), \dots, V_n^{i,s}(t)) \tag{2}$$

collects these quantities over the first n ranks.

Raw displayed volume may not be comparable across assets or intraday periods with different market depth. The reported implementation therefore divides participant volume by total displayed market depth at the same side and rank:

$$\tilde{V}_k^{i,s}(t) = \frac{V_k^{i,s}(t)}{D_k^s(t)}, \tag{3}$$

where $D_k^s(t)$ is the total displayed market depth on side s and rank k . A rank with zero market depth contributes zero; an imbalance is undefined when the participant has no positive weighted

footprint on either side. We denote the distance from the touch by $d_k^s(t)$. In tick units, this distance equals the number of ticks between rank k and the same-side best quote, shifted upward by one. The shift gives Rank 1 a positive distance and preserves the ordering of deeper ranks.

5.1.2 Multilevel Distance-Weighted Imbalance

Economic intuition and a stylized parametric benchmark. We use a depth kernel to reduce the top- n vector to a scalar weighted-liquidity measure. A high-weight rank should remain visible to traders who process multilevel data but face relatively low immediate execution risk. The following parametric kernel illustrates this trade-off:

$$\omega_k^{s,\text{par}}(t) = \frac{e^{-\lambda d_k^s(t)} (1 - e^{-\kappa d_k^s(t)})}{\sum_{\ell=1}^n e^{-\lambda d_\ell^s(t)} (1 - e^{-\kappa d_\ell^s(t)})}, \quad (4)$$

The visibility component $e^{-\lambda d}$ decreases with depth. The protection component $1 - e^{-\kappa d}$ increases as the displayed price moves away from immediate execution. Their product penalizes both the high execution risk near the touch and the lower visibility deeper in the book. The parameters λ and κ control the two shapes. This expression is only a conceptual benchmark: the empirical analysis uses neither these weights nor fitted values of λ and κ . Instead, it constructs rank-level empirical proxies for each instrument and book side.

Figure 3 illustrates the benchmark trade-off and its discrete normalization; it is not an estimated empirical profile.

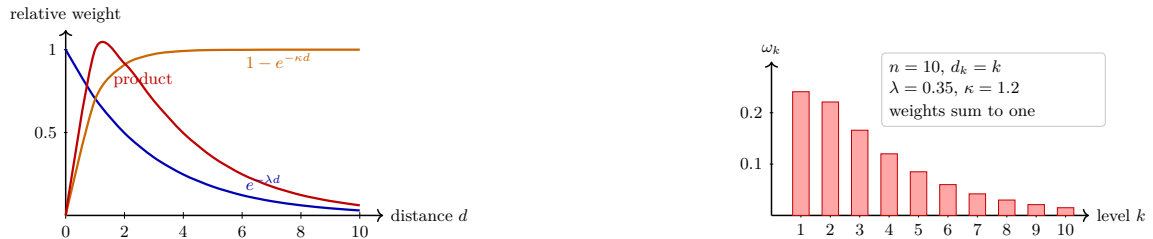


Figure 3: Conceptual depth kernel in equation (4). The left panel shows decreasing visibility, increasing protection from immediate execution, and their product. The right panel shows normalized weights for one illustrative parameter choice. The empirical analysis uses the instrument- and side-specific kernel in equation (11), not these parametric weights.

Operational empirical depth kernel. The operational kernel replaces both exponential components with empirical rank-level quantities. We calibrate each book rank k separately for instrument m and side s . A rank receives more weight when its depth has a strong short-horizon association with mid-price changes and its displayed price is rarely reached over the same horizon. Tick distance can vary for a given rank over time, but the calibrated unit remains the rank.

Operational calculation. We reconstruct a post-event top- n snapshot after every ordered message. Let $\mathcal{O}_{m,s,k}$ contain the snapshots in which rank k exists on side s and has positive displayed depth. For each $t \in \mathcal{O}_{m,s,k}$, the indicator $R_{t,k}^{m,s}(H)$ equals one when at least one fill during $(t, t + H]$ occurs at or through the rank's displayed price. An ask rank is reached when

the maximum ask fill price in the window is at least the displayed price. A bid rank is reached when the minimum bid fill price is at most the displayed price. We estimate the rank-level hit probability from pooled hit and exposure counts:

$$\hat{p}_k^{m,s}(H) = \frac{\sum_{t \in \mathcal{O}_{m,s,k}} R_{t,k}^{m,s}(H)}{|\mathcal{O}_{m,s,k}|}. \quad (5)$$

Each observable snapshot contributes once, even when the same rank appears at different tick distances. Thus, [equation \(5\)](#) divides total hits by total exposures; it does not average distance-specific hit rates. Realized tick distance is retained only as descriptive metadata and does not define additional kernel rows or enter a parametric fit. The estimator is a short-horizon proxy for whether the displayed price is reached. It is not an individual order's fill probability, which also depends on queue position, order size, and intervening cancellations. We define the protection component as the complement of the hit probability:

$$\hat{\rho}_k^{m,s}(H) = 1 - \hat{p}_k^{m,s}(H), \quad (6)$$

A value of $\hat{\rho}_k^{m,s}(H)$ near one indicates that rank k is rarely reached over horizon H .

The second component measures the association between displayed depth and subsequent short-horizon price changes. Let $D_k^{m,s}(t)$ denote total market depth, rather than participant- i depth, at rank k . Total displayed top- n depth at snapshot t is

$$D_n^m(t) = \sum_{r=1}^n \left(D_r^{m,a}(t) + D_r^{m,b}(t) \right). \quad (7)$$

We then define the signed and normalized contribution of side s and rank k :

$$Z_{t,k}^{m,s} = \sigma_s \frac{D_k^{m,s}(t)}{D_n^m(t)}, \quad \sigma_s = \begin{cases} +1, & s = ask, \\ -1, & s = bid. \end{cases} \quad (8)$$

The magnitude $|Z_{t,k}^{m,s}|$ is the fraction of visible top- n depth supplied by that side and rank. The sign makes ask and bid depth contribute in opposite directions to imbalance.

For each snapshot, $M_{t,H}^m$ is the last finite reconstructed mid-price observed after t and no later than $t + H$. The corresponding price change is $\Delta M_{t,H}^m = M_{t,H}^m - M^m(t)$. Let $\mathcal{V}_{m,s,k}(H)$ contain the rank- k snapshots for which both mid-prices are finite. For each instrument–side–rank cell, we compute the following pooled population covariance in absolute value, where the overbars denote sample means over $\mathcal{V}_{m,s,k}(H)$:

$$\hat{\nu}_k^{m,s}(H) = \left| \frac{1}{|\mathcal{V}_{m,s,k}(H)|} \sum_{t \in \mathcal{V}_{m,s,k}(H)} \left(Z_{t,k}^{m,s} - \overline{Z}_k^{m,s} \right) \left(\Delta M_{t,H}^m - \overline{\Delta M}_{k,H}^{m,s} \right) \right|. \quad (9)$$

The absolute value makes $\hat{\nu}_k^{m,s}(H)$ a measure of the magnitude, rather than the direction, of rank-specific co-movement. We pool observations across realized tick distances before computing the covariance. We do not compute or average separate distance-bucket covariances. This quantity is a weighting proxy, not a causal estimate of price impact. It is also expressed in

instrument-specific price units. We therefore use it only to allocate weights across ranks on the same side of one instrument, not to compare covariance magnitudes across instruments.

The empirical kernel follows the same multiplicative decomposition as [equation \(4\)](#), but no exponential curve is fitted. Before normalization, we apply a zero floor to protection and a 10^{-12} floor to the absolute covariance:

$$\hat{\rho}_k^{m,s}(H) = \max\{1 - \hat{p}_k^{m,s}(H), 0\}, \quad \tilde{\nu}_k^{m,s}(H) = \max\{\hat{\nu}_k^{m,s}(H), 10^{-12}\}, \quad (10)$$

We normalize the product of these two quantities within each instrument and side:

$$\hat{\omega}_k^{m,s}(H) = \frac{\tilde{\nu}_k^{m,s}(H) \hat{\rho}_k^{m,s}(H)}{\sum_{\ell=1}^n \tilde{\nu}_\ell^{m,s}(H) \hat{\rho}_\ell^{m,s}(H)}. \quad (11)$$

The denominator must be finite and strictly positive. If every rank on a side has zero protection, or if the product is otherwise degenerate, calibration fails instead of returning an uninterpretable all-zero vector. Every reported kernel in [table 2](#) passes this condition and sums to one before rounding. A rank therefore receives high weight only when it combines price association with low short-horizon execution risk.

In the remaining formulas, the operational value of $\omega_k^s(t)$ is $\hat{\omega}_k^{m,s}(H)$. Every event for instrument m uses the same calibrated kernel. The kernel is not re-estimated by actor, event, or observed outcome. Calibration produces one row for each (m, s, k) cell, and the runtime loader rejects duplicate rank rows instead of resolving them by row order. The descriptive analysis uses the full available sample for each instrument. A prospective or out-of-sample application must instead estimate the kernel on an earlier training period and hold it fixed, so that future order flow does not enter the weights.

Table 2: Operational full-sample empirical top-10 depth-kernel weights at horizon $H = 10$ seconds. Values are rounded to three decimal places and sum to one within each instrument and side before rounding. These weights, rather than fitted exponential parameters, enter the empirical spoofing metrics.

Instrument	Side	$\hat{\omega}_1$	$\hat{\omega}_2$	$\hat{\omega}_3$	$\hat{\omega}_4$	$\hat{\omega}_5$	$\hat{\omega}_6$	$\hat{\omega}_7$	$\hat{\omega}_8$	$\hat{\omega}_9$	$\hat{\omega}_{10}$
Risanamento	Ask	0.127	0.248	0.348	0.062	0.004	0.039	0.018	0.012	0.078	0.064
	Bid	0.526	0.012	0.019	0.146	0.091	0.065	0.082	0.034	0.002	0.024
Nexi	Ask	0.028	0.069	0.033	0.138	0.225	0.149	0.275	0.011	0.020	0.051
	Bid	0.045	0.166	0.184	0.276	0.165	0.033	0.006	0.052	0.064	0.008
Ferrari	Ask	0.095	0.022	0.016	0.209	0.270	0.032	0.144	0.032	0.033	0.147
	Bid	0.148	0.107	0.007	0.054	0.072	0.197	0.102	0.038	0.176	0.100

The operational calibration uses the first ten ranks and the same $H = 10$ second horizon for every instrument. It produces 20 weights per instrument, one for each side–rank cell. The kernel remains side-specific. To draw one curve per instrument for visualization only, we average the bid and ask weights at each rank:

$$\bar{\omega}_k^m(H) = \frac{1}{2} \left(\hat{\omega}_k^{m,a}(H) + \hat{\omega}_k^{m,b}(H) \right). \quad (12)$$

Each side-specific kernel sums to one, so $\sum_{k=1}^{10} \bar{\omega}_k^m(H) = 1$. We use this average only in [figure 4](#). The empirical surveillance metrics use the side-specific top-10 weights in [table 2](#).

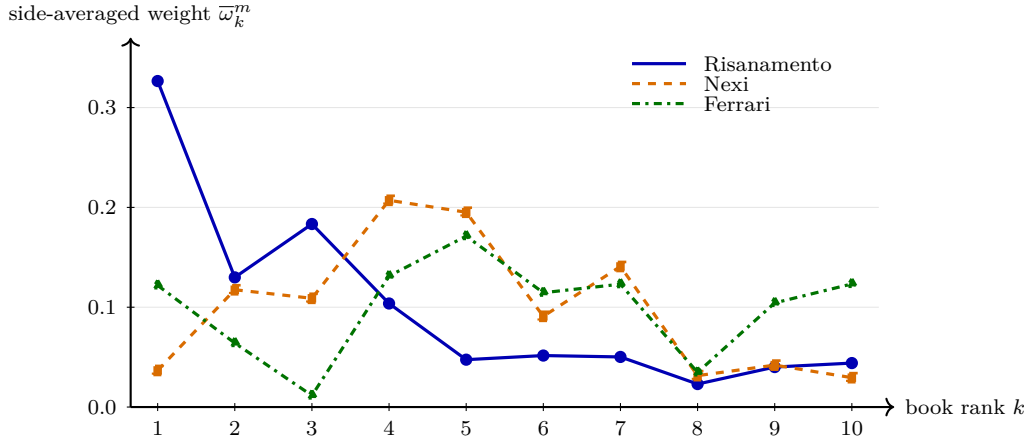


Figure 4: Full-sample empirical top-10 depth-kernel profiles at $H = 10$ seconds. Each line is the arithmetic mean of the separately normalized bid and ask weights in [equation \(12\)](#). The mean permits one curve per instrument on a common rank axis and is used only for visualization. Surveillance metrics use the underlying side-specific kernels.

To expose the two empirical inputs rather than only their normalized product, [Figures 5](#) and [6](#) plot the side-specific visibility and execution-risk profiles used in the calibration. The second plot reports $p_k^{i,s}$ directly; the kernel uses its protection complement $\rho_k^{i,s} = 1 - p_k^{i,s}$.

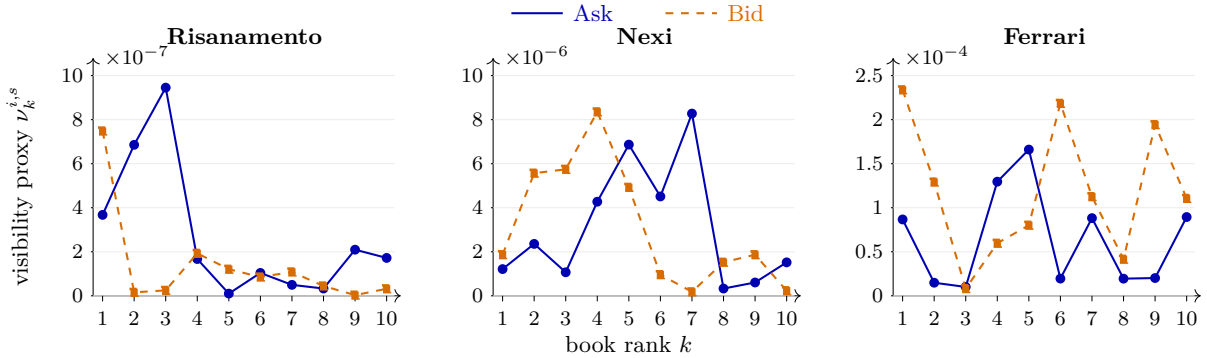


Figure 5: Side-specific empirical visibility proxy by book rank. Each panel shows the stored `visibility_component` values from the full-sample top-10, $h = 10$ -second calibration. Separate axis multipliers preserve the instrument-specific price units; vertical magnitudes should not be compared across instruments.

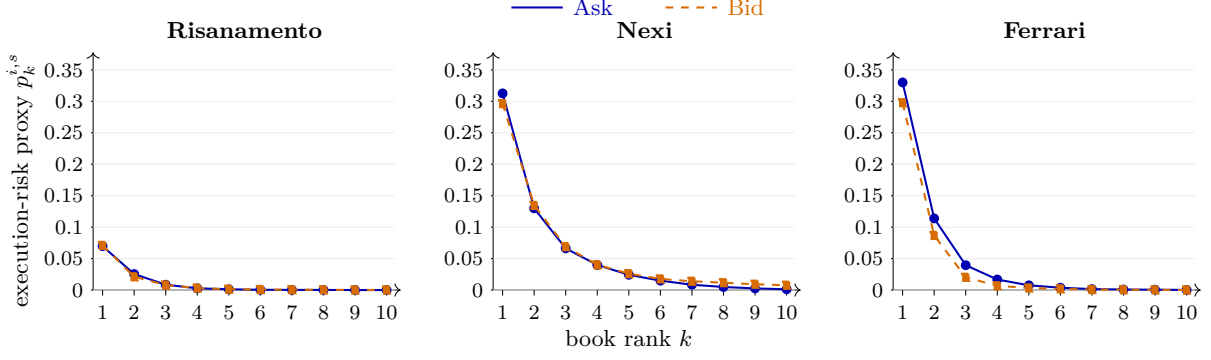


Figure 6: Side-specific empirical execution-risk proxy by book rank. The plotted quantity is the stored h -second reach probability $p_k^{i,s}$ (`hit_probability`) from the full-sample top-10, $h = 10$ -second calibration. The empirical kernel uses its protection complement $\rho_k^{i,s} = 1 - p_k^{i,s}$.

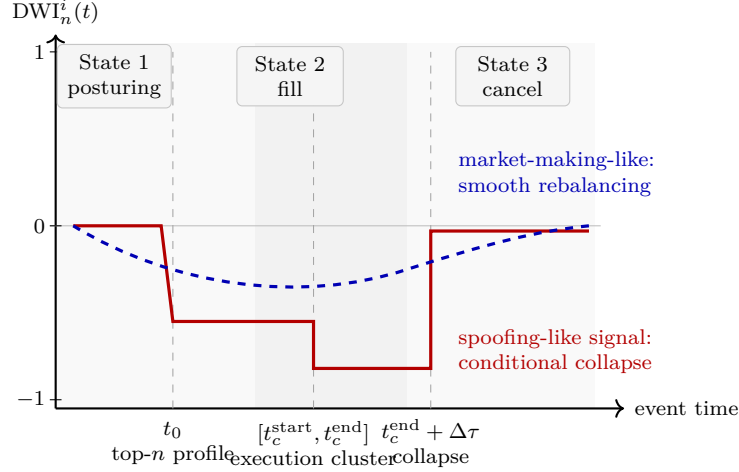
Using the calibrated kernel, we define participant i 's weighted top- n liquidity on side s as

$$L_n^{i,s}(t) = \sum_{k=1}^n \omega_k^s(t) \tilde{V}_k^{i,s}(t). \quad (13)$$

We summarize the difference between the two sides with the *Multilevel Distance-Weighted Imbalance*:

$$\text{DWI}_n^i(t) = \frac{L_n^{i,a}(t) - L_n^{i,b}(t)}{L_n^{i,a}(t) + L_n^{i,b}(t)}. \quad (14)$$

When the denominator is positive, $\text{DWI}_n^i(t) > 0$ denotes an ask-heavy profile and $\text{DWI}_n^i(t) < 0$ denotes a bid-heavy profile. The denominator scales the difference by the participant's total weighted footprint, reducing sensitivity to the absolute scale of that footprint within the same instrument and kernel calibration. Setting $n = 1$ gives the best-level measure. Larger values of n also capture candidate liquidity posted behind the touch. Figure 7 illustrates two stylized paths; it is not an empirical classification rule.



For the opposite spoofing direction, the sign pattern is mirrored.

Figure 7: Stylized cluster-time behavior of the multilevel distance-weighted imbalance. In the spoofing-compatible pattern, a large top- n imbalance returns rapidly toward zero after the smaller opposite-side execution because the candidate profile is cancelled. The illustrative market-making path shows smoother inventory rebalancing.

5.1.3 Conditional Withdrawal Reconstruction

A static multilevel imbalance cannot establish manipulative intent. A market maker can also become temporarily imbalanced while controlling inventory or responding to adverse selection. The additional observable feature is whether the profile disappears conditionally after the participant's execution on the opposite side, as predicted by the stylized mechanism.

Let c denote an execution cluster for canonical actor i . Its ordered child executions share the trading partition, actor key, order identifier, execution role, and side, and they satisfy the operational inter-fill-gap rule. Passive child executions also share the resting price. Aggressive child executions use reported traded price and quantity and may span several price levels within the same order or sweep. The timestamps t_c^{start} and t_c^{end} denote the first and last child executions. The quantity Q_c is their total executed quantity, and $s_c \in \{b, a\}$ is the actor's execution side. We denote the opposite candidate-posture side by $\bar{s}_c$. The pre- and post-cluster observation times are

$$t_c^- = t_c^{\text{start}} - \epsilon_t, \quad t_c^+ = t_c^{\text{end}} + \Delta\tau, \quad (15)$$

where ϵ_t places t_c^- immediately before the first child fill and $\Delta\tau$ bounds the withdrawal window after the final child fill.

We also impose a pre-execution episode window $\Theta > 0$. An opposite-side order enters the candidate profile for cluster c only when its first observed posting time τ_o satisfies

$$0 \leq t_c^{\text{start}} - \tau_o \leq \Theta. \quad (16)$$

The empirical analysis uses $\Theta = 90$ seconds as an exploratory default. Episode attribution therefore includes only same-actor top- n orders first observed inside this pre-execution window. This restriction separates a recent candidate layer from the participant's standing background liquidity. Without it, a nearly balanced long-lived profile could dilute the measured withdrawal

of a short-lived recent layer. We retain the unrestricted top- n state to describe the ordinary book footprint and use an *episode-conditioned* profile for surveillance attribution.

Let $\mathcal{A}_{c,k,\Theta}^{i,s}(t)$ contain the active orders of actor i on side s and rank $k \leq n$ at time t whose first observed posting time lies in $[t_c^{\text{start}} - \Theta, t_c^{\text{start}}]$. Their episode-conditioned weighted liquidity is

$$L_{c,n,\Theta}^{i,s}(t) = \sum_{k=1}^n \omega_k^s(t) \sum_{o \in \mathcal{A}_{c,k,\Theta}^{i,s}(t)} \tilde{q}_o(t), \quad (17)$$

where $\tilde{q}_o(t)$ is the visible quantity of order o , optionally normalized by the corresponding market depth as in [equation \(3\)](#). Write $T_{c,n,\Theta}^i(t) = L_{c,n,\Theta}^{i,a}(t) + L_{c,n,\Theta}^{i,b}(t)$ for the corresponding total weighted liquidity. We define the episode-conditioned imbalance piecewise as

$$\text{DWI}_{c,n,\Theta}^i(t) = \begin{cases} \frac{L_{c,n,\Theta}^{i,a}(t) - L_{c,n,\Theta}^{i,b}(t)}{T_{c,n,\Theta}^i(t)}, & T_{c,n,\Theta}^i(t) > 0, \\ 0, & T_{c,n,\Theta}^i(t) = 0 \text{ and the zero-profile state is retained,} \\ \text{undefined,} & \text{otherwise.} \end{cases} \quad (18)$$

This statistic describes the recent liquidity layer associated with cluster c , rather than the actor's entire standing book. Active profiles with a zero weighted denominator remain undefined rather than being converted into finite evidence. All subsequent collapse and withdrawal diagnostics use this episode-conditioned profile.

The *Spoofing Conditionality Index* measures the absolute change in its normalized shape:

$$\text{SCI}_{c,n}^i = \left| \text{DWI}_{c,n,\Theta}^i(t_c^-) - \text{DWI}_{c,n,\Theta}^i(t_c^+) \right|. \quad (19)$$

An absolute imbalance change does not identify which side lost liquidity. We therefore also measure the fraction of weighted liquidity that disappears from each side after the fill using an exact piecewise definition:

$$C_{c,n}^{i,s} = \begin{cases} \frac{[L_{c,n,\Theta}^{i,s}(t_c^-) - L_{c,n,\Theta}^{i,s}(t_c^+)]_+}{L_{c,n,\Theta}^{i,s}(t_c^-)}, & L_{c,n,\Theta}^{i,s}(t_c^-) > 0, \\ 0, & L_{c,n,\Theta}^{i,s}(t_c^-) = 0. \end{cases} \quad s \in \{b, a\}. \quad (20)$$

where $[x]_+ = \max\{x, 0\}$. A large $C_{c,n}^{i,\bar{s}_c}$ indicates that recent liquidity disappeared from the side opposite to the clustered execution, as required by the mechanical sequence. A two-sided quote refresh after a mid-price change can instead generate material collapse on both sides.

To summarize these diagnostics while preserving evidence in both directions, define the event-level signed contrast

$$\text{MSCI}_{c,n}^i = \frac{1}{2} \text{SCI}_{c,n}^i + C_{c,n}^{i,\bar{s}_c} - C_{c,n}^{i,s_c}. \quad (21)$$

Because DWI lies in $[-1, 1]$, $\text{SCI}/2$ and each collapse term lie in $[0, 1]$. Therefore, MSCI lies in $[-1, 2]$ without an additional clipping step. A positive MSCI means that normalized shape change and opposite-side collapse outweigh same-side collapse. Zero is the neutral contrast,

whereas a negative MSCI means that same-side collapse dominates.

The positive part in the definition of C distinguishes liquidity disappearance from liquidity growth. It does not truncate the subsequent cross-side contrast. The three inputs remain separately inspectable, so a two-sided quote refresh is distinguishable from an asymmetric collapse. MSCI is a secondary event-level diagnostic that enters actor-level feature summaries. It is neither a probability nor matched-withdrawal evidence. It does not enter the four-condition event gate or the generated result tables.

5.1.4 Canonical Matched Withdrawal

A cancellation can be eligible for several nearby execution clusters. To measure withdrawal without counting that cancellation more than once, let $\mathcal{E}(o)$ contain the eligible same-actor, opposite-side clusters that precede cancellation o by at most $\Delta\tau$. We assign o deterministically to the canonical cluster

$$a(o) = \arg \min_{c \in \mathcal{E}(o)} (t_o - t_c^{\text{end}}, t_c^{\text{start}}, \text{id}_c), \quad (22)$$

where the tuple is ordered lexicographically and the stable cluster identifier breaks any final tie. Eligibility also requires source-sequence precedence: the cancellation follows the cluster's final child fill in both event time and source sequence. For eligible top- n cancellations, define $\mathcal{C}_{c,n}^i = \{o : a(o) = c\}$. A physical cancellation is identified by its trading partition, source index, and candidate order identifier. We preserve every eligible link for audit, but each physical cancellation contributes to at most one cluster's metrics.

For each assigned order $o \in \mathcal{C}_{c,n}^i$, let q_o denote its visible quantity immediately before cancellation. The total assigned withdrawal and its ratio to cluster quantity are

$$U_{c,n}^i = \sum_{o \in \mathcal{C}_{c,n}^i} q_o, \quad R_{c,n}^i = \frac{U_{c,n}^i}{Q_c}, \quad (23)$$

where $Q_c > 0$ is the execution-cluster quantity by construction. Let $Q_{c,n}^{i,\bar{s}_c}$ denote the visible quantity of the candidate profile before execution. We report the fraction of that profile removed as

$$A_{c,n}^i = \begin{cases} \frac{U_{c,n}^i}{Q_{c,n}^{i,\bar{s}_c}}, & Q_{c,n}^{i,\bar{s}_c} > 0, \\ \text{undefined}, & Q_{c,n}^{i,\bar{s}_c} = 0. \end{cases} \quad (24)$$

These quantities remain separately inspectable. The full descriptive gate requires $U_{c,n}^i > Q_c$ rather than applying a threshold to a composite score. We summarize repeated behavior through counts and withdrawal rates at the actor-partition level. We do not use a scalar aggregate to rank actors or infer intent.

5.1.5 Price-Response Diagnostics

The matched-withdrawal variables show whether an actor removes same-actor opposite-side liquidity after its own execution. They do not show whether the price path is consistent with the hypothesized mechanism. Such a path first moves in the direction favorable to the clustered

execution and then partly reverses after the candidate layer is removed. We measure these two features with separate price-response diagnostics. They are economic consistency checks, not causal estimates of manipulation.

Let $b(t)$ and $a(t)$ denote the best bid and ask, with corresponding displayed quantities $q_1^b(t)$ and $q_1^a(t)$. When both quotes and positive top-level depth are available, the two market-price proxies are the mid-price $M_{\text{mid}}(t) = [a(t) + b(t)]/2$ and the queue microprice $M_{\text{micro}}(t) = [a(t)q_1^b(t) + b(t)q_1^a(t)]/[q_1^b(t) + q_1^a(t)]$ [12]; otherwise the corresponding diagnostic is missing. Let $M(t)$ denote either proxy, and let t_0 be the earliest first-observed posting time among the eligible candidate orders for cluster c . We encode the clustered execution direction as

$$h_c = \begin{cases} +1, & \text{if the clustered execution is a sell execution,} \\ -1, & \text{if the clustered execution is a buy execution.} \end{cases} \quad (25)$$

Under this convention, a positive signed change is favorable to the execution: a sell benefits from an upward move and a buy from a downward move. We measure the favorable movement between candidate-layer posting and execution as

$$\text{FPM}_c^i = h_c [M(t_c^-) - M(t_0)], \quad (26)$$

Reversion is anchored to each assigned cancellation rather than to the end of the execution cluster. Let t_o denote the cancellation timestamp, t_o^- the last reconstructed state strictly before its source index, and $H_R = 2$ seconds the reversion horizon. Let $\mathcal{C}_{c,n}^{i,\text{obs}}$ contain the assigned cancellations with both pre-cancellation and horizon states. The cluster-level post-cancellation reversion is the quantity-and-delay weighted mean

$$\text{REV}_c^i = \frac{\sum_{o \in \mathcal{C}_{c,n}^{i,\text{obs}}} w_{c,o} h_c [M(t_o^-) - M(t_o + H_R)]}{\sum_{o \in \mathcal{C}_{c,n}^{i,\text{obs}}} w_{c,o}}, \quad w_{c,o} = q_o \exp\left(-\frac{t_o - t_c^{\text{end}}}{\tau_w}\right), \quad (27)$$

with $\tau_w = 10$ seconds in the reported run. Here $M(t_o + H_R)$ denotes the last reconstructed state at or before the target time, provided that the message sequence covers that target. The statistic is missing when no assigned cancellation has an observable horizon state. A positive FPM_c^i means that the price moved in the direction favorable to the clustered execution while the candidate layer was present. A positive REV_c^i means that the market-price proxy moved against the pre-execution favorable direction after the assigned cancellation or cancellations. For quantity-weighted cluster price P_c , we also define the local execution advantage relative to the pre-posture benchmark:

$$\text{ADV}_c^i = h_c [P_c - M(t_0)]. \quad (28)$$

The execution advantage is positive when a sell executes above the benchmark or a buy executes below it.

We compute each diagnostic with both the mid-price and the queue microprice in the event-level artifacts. The generated summary tables and the event gate use the mid-price versions. We report price responses separately from withdrawal counts, quantities, and rates instead of multiplying all components into one detector. Withdrawal variables describe the reconstructed

message sequence. The variables FPM, REV, and ADV describe the associated price path. This separation distinguishes repeated conditional withdrawals without a favorable price response from episodes that also exhibit favorable pre-execution movement and post-cancellation reversion.

5.1.6 Event Reconstruction

We reconstruct events from order-level messages rather than aggregate snapshots. A resting order can change book rank when the best quote moves even though its owner has neither modified nor cancelled it. Snapshot changes alone can therefore confound repricing of the surrounding book with withdrawal by the participant. The operational procedure is:

1. Resolve the canonical actor and classify each fill as exclusively passive or aggressive. Construct cluster c from compatible child executions while preserving the ordered mapping to raw messages.
2. Reconstruct participant i 's active orders immediately before t_c^{start} .
3. Assign each active order to side s and rank $k \leq n$. Retain in the episode-conditioned profile only orders whose first observed posting time lies in $[t_c^{\text{start}} - \Theta, t_c^{\text{start}}]$.
4. Compute $L_{c,n,\Theta}^{i,b}(t_c^-)$, $L_{c,n,\Theta}^{i,a}(t_c^-)$, and $\text{DWI}_{c,n,\Theta}^i(t_c^-)$.
5. Apply executions, cancellations, and amendments through $[t_c^{\text{start}}, t_c^{\text{end}} + \Delta\tau]$.
6. Retain every eligible cluster–cancellation link. Assign each physical cancellation canonically using [equation \(22\)](#), and exclude competing links from metric totals.
7. Compute the corresponding episode-conditioned quantities at t_c^+ .
8. Report assigned withdrawal quantity, withdrawal-to-execution ratio, and removed-profile fraction.
9. Compute signed price diagnostics from the first candidate-layer posting, through the cluster, to the post-cancellation window.
10. Summarize repeated behavior with cluster counts and actor-level recurrence diagnostics. Retain raw child-execution messages as provenance.

This procedure is designed to distinguish ordinary book repricing from withdrawal of participant-attributed candidate liquidity. Canonical assignment ensures that one physical cancellation does not inflate the metrics of several neighboring clusters under the stated eligibility rule.

The event-level *spoofing-compatible sequence* gate requires four observed conditions: a rapid assigned withdrawal, withdrawn quantity greater than cluster execution quantity, positive favorable pre-execution mid-price movement, and positive mid-price reversion from the assigned cancellation over an observable 2-second horizon. Right-censored reversion outcomes fail the gate. This transparent conjunction supports analyst triage; it is descriptive and is not a statistical test, manipulation label, or proof of intent.

The all-sample run uses a 10-second structural window, 2-second rapid-attribution and cancel-anchored reversion windows, and a 90-second candidate-order age limit. We estimate the empirical top-10 kernel separately for each instrument. The reported outputs keep withdrawal and price-response components separate rather than reducing them to a scalar composite score.

6 Results

6.1 Cross-Instrument Detection Summary

We apply the same episode-conditioned top-10 reconstruction to all three samples. Table 3 reports the resulting exploratory diagnostics. Each instrument uses its full-sample empirical kernel from table 2 with $H = 10$ seconds.

A matched cluster is a passive or aggressive execution followed within 2 seconds by a canonically assigned cancellation of a same-actor, opposite-side candidate order. The separate 10-second interval is the structural reconstruction horizon. The table separates the two execution anchors, distinguishes clusters from their raw child execution messages, and counts each assigned physical cancellation once. Its firm-fallback column shows how many matched clusters use firm identity because the original client code is missing. The strict column counts clusters that also satisfy the withdrawal-size, favorable mid-price movement, and post-cancellation reversion conditions. These quantities are descriptive screening diagnostics, not manipulation labels or evidence of intent.

Table 3: Actor-attributed execution-cluster populations and reconstructed withdrawals by execution anchor. Raw fills are provenance messages; clusters are the analytical unit. Firm fallback counts matched clusters for which the client code is missing and the firm identifier defines the actor. Assigned cancellations are counted once. Strict is the four-condition spoofing-compatible sequence gate.

Instrument	Anchor	Clusters	Raw fills	Matched	Firm fallback	Assigned cancels	Strict
FERRARI	Aggressive	4,333	9,335	1,247	1,036	1,922	175
FERRARI	Passive	8,664	9,329	1,336	1,098	2,034	203
NEXI	Aggressive	3,849	7,425	621	444	945	100
NEXI	Passive	6,663	7,424	1,064	711	1,868	131
RISANAMENTO	Aggressive	9,956	16,029	71	14	84	10
RISANAMENTO	Passive	15,016	15,929	134	92	136	2

The reported empirical quantities are direct reconstruction counts and conditional descriptive shares. No counterfactual comparison or inferential test is applied, so the table does not provide p -values, confidence intervals, or evidence that withdrawal is more likely after the focal actor’s own execution than under an otherwise comparable no-execution condition.

6.2 Risanamento

The Risanamento sample contains 31,958 raw execution messages, consolidated into 24,972 clusters: 15,016 passive and 9,956 aggressive. Canonical linkage identifies 205 matched clusters, including 134 passive and 71 aggressive anchors. 106 matched clusters use firm fallback. These clusters own 220 physical cancellations, each assigned at most once, and 12 satisfy the strict

four-condition gate. Among matched clusters with an observed diagnostic, 56.6% have positive favorable pre-execution movement and 21.7% have positive cancel-anchored reversion.

The event-review dashboard contains a cluster-first dossier and reconstructed top-10 window for each of the 205 matched clusters. These review objects organize observed facts, counter-evidence, and legitimate alternatives. They are triage aids, not empirical labels or proof of manipulation.

6.3 Nexi

The Nexi sample contains 14,849 raw execution messages and 10,512 clusters, split into 6,663 passive and 3,849 aggressive anchors. Reconstruction identifies 1,685 matched clusters; 1,155 use firm fallback. They own 2,813 assigned cancellations, and 231 pass the strict gate. Positive favorable pre-execution movement occurs in 52.7% of matched clusters with an observed diagnostic, and positive cancel-anchored reversion occurs in 36.5%.

Differences from the other instruments are not causal contrasts because the samples cover different periods and actor compositions.

6.4 Ferrari

The Ferrari sample contains 18,664 raw execution messages and 12,997 clusters, of which 8,664 are passive and 4,333 are aggressive. Reconstruction identifies 2,583 matched clusters; 2,134 use firm fallback. They own 3,956 assigned cancellations, and 378 pass the strict gate. The shares with positive favorable pre-execution movement and positive reversion among observed matched-cluster diagnostics are 54.1% and 35.7%.

These are descriptive screening outcomes, not manipulation labels or prevalence estimates.

6.5 Actor-Level Attribution and Interpretation

Table 4: Largest actor–anchor rapid-withdrawal groups by matched-cluster count. Client denotes the original client code. Firm fallback denotes aggregation by firm only when that client code is missing. W/E max is the largest assigned withdrawal-to-execution ratio in the group. FPM+ and REV+ are descriptive shares among matched clusters with finite mid-price diagnostics.

Instrument	Actor	Identity	Anchor	Matched	W/E max	FPM+	REV+
FERRARI	157922_3	Firm fallback	Aggressive	367	60.0	43.6%	32.7%
FERRARI	157922_3	Firm fallback	Passive	257	77.5	47.1%	30.4%
FERRARI	619575_1	Firm fallback	Passive	242	34.5	52.1%	41.7%
NEXI	23523	Client	Passive	248	1200.0	64.5%	25.4%
NEXI	354310	Firm fallback	Passive	233	565.2	59.7%	28.9%
NEXI	628912	Firm fallback	Aggressive	191	1000.0	33.0%	57.1%
RISANAMENTO	628912	Firm fallback	Passive	83	80.0	79.5%	1.2%
RISANAMENTO	0	Client	Aggressive	30	26.8	36.7%	56.7%
RISANAMENTO	0	Client	Passive	22	6.5	77.3%	13.6%

Table 4 reports the three largest actor–anchor groups in each instrument by matched-cluster count. An actor is either an original client code or, only when that code is missing, a firm identifier. The identity and anchor columns keep these dimensions explicit. W/E maximum is the largest matched withdrawal-to-execution ratio within the group; FPM+ and REV+ are the

shares of matched clusters with positive mid-price favorable movement and reversion. Because firm fallback can combine several clients, a firm-level count has coarser attribution than a client-level count and should not be interpreted as a single-account history.

The primary empirical object is the participant-level association between execution and rapid opposite-side withdrawal. We report its magnitude and repetition descriptively. An analyst must also consider the withdrawn order’s execution risk and distance from the touch, bilateral activity, inventory context, price response, and execution advantage. Together, these diagnostics indicate whether an episode is economically consistent with the stylized mechanism or remains compatible with legitimate liquidity provision. Neither a scalar score nor a dashboard review category is treated as ground truth.

7 Discussion and Conclusion

Across these three samples, the reconstruction identifies rapid same-actor withdrawal episodes after both passive and aggressive executions, together with their associated price paths. These outcomes describe the stated reconstruction and thresholds; they neither estimate spoofing prevalence nor establish that any retained sequence was manipulative.

Several limitations constrain interpretation. The depth kernels are calibrated on the same full samples to which they are applied, so their displayed-price associations are descriptive rather than out-of-sample evidence. The analysis does not include a counterfactual statistical test, and repeated events from the same actors or trading periods need not be independent. Right-censored reversion outcomes fail the event gate, and the reported results do not test sensitivity to the 100-millisecond child-execution clustering threshold. Firm fallback loses within-firm client resolution and can pool several accounts, so recurrence at that level is not directly comparable with recurrence under a client code. Aggressive execution is an execution-role observation, not evidence about the intent of the surrounding resting orders; ordinary liquidity taking remains an alternative explanation. The three instrument samples also cover different periods and actor compositions, which rules out causal cross-instrument comparisons.

The contribution is therefore an auditable surveillance workflow rather than an enforcement classifier. It preserves raw message provenance, separates withdrawal from price-response evidence, and exposes the gate conditions used for triage. Prospective use would require a pre-specified counterfactual design and thresholds, kernel calibration on training periods only, chronological out-of-sample validation, and evaluation against independently reviewed cases with explicit false-positive and false-negative costs.

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